



Derivatives of Trigonometric Functions

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Apply derivative formulas for the six trigonometric functions
- Use the product and quotient rules with trigonometric functions
- Simplify derivative expressions by factoring common terms

Find the derivative of each function and simplify your answer completely.

1. Find the derivative of the function.

$$y = 2\csc x + 5\cos x$$

Answer: _____

2. Find the derivative using the product rule.

$$y = e^x \tan x$$

Answer: _____

3. Find the derivative using the quotient rule.

$$y = \frac{\sin x + 1}{-3\cos x}$$

Answer: _____

4. Find the derivative of the function.

$$y = 3\sin x - 4\sec x$$

Answer: _____

5. Find the derivative using the product rule.

$$y = x^2 \cos x$$

Answer: _____

6. Find the derivative of the function.

$$y = \cot x + 7\tan x$$

Answer: _____

7. Find the derivative using the quotient rule.

$$y = \frac{\cos x}{x}$$

Answer: _____



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8. Find the derivative using the product rule.

$$y = \sin x \cos x$$

Answer: _____

9. Find the derivative of the function.

$$y = 4\csc x - 2\cot x$$

Answer: _____

10. Find the derivative using the product rule.

$$y = e^x \sin x$$

Answer: _____





Students should memorize the six trig derivative formulas before attempting the product and quotient rule problems.

Solutions

1. Find the derivative of the function.

$$y = 2\csc x + 5\cos x$$

- Apply the constant multiple rule to each term
- The derivative of cosecant x is negative cosecant x times cotangent x
- The derivative of cosine x is negative sine x
- Multiply by the constants and combine the results

Answer: $y' = -2\csc x \cot x - 5\sin x$

2. Find the derivative using the product rule.

$$y = e^x \tan x$$

- Identify the two factors as e to the x and tangent x
- Apply the product rule: derivative of first times second plus first times derivative of second
- The derivative of e to the x is e to the x and the derivative of tangent x is secant squared x
- Factor out the greatest common factor e to the x to simplify

Answer: $y' = e^x(\tan x + \sec^2 x)$

3. Find the derivative using the quotient rule.

$$y = \frac{\sin x + 1}{-3\cos x}$$

- Identify the numerator as sine x plus one and the denominator as negative three cosine x
- Apply the quotient rule: low times derivative of high minus high times derivative of low all over low squared
- The derivative of sine x is cosine x and the derivative of negative three cosine x is three sine x
- Expand the numerator and simplify using the Pythagorean identity if possible

Answer: $y' = \frac{-3 - 3\sin x}{-3\cos^2 x}$

4. Find the derivative of the function.

$$y = 3\sin x - 4\sec x$$

- Take the derivative of each term separately using the constant multiple rule
- The derivative of sine x is cosine x
- The derivative of secant x is secant x times tangent x
- Combine the terms with their constants

Answer: $y' = 3\cos x - 4\sec x \tan x$



5. Find the derivative using the product rule.

$$y = x^2 \cos x$$

- Identify the two factors as x squared and cosine x
- Apply the product rule formula
- The derivative of x squared is two x and the derivative of cosine x is negative sine x
- Write out the resulting expression

Answer: $y' = 2x \cos x - x^2 \sin x$

6. Find the derivative of the function.

$$y = \cot x + 7 \tan x$$

- Differentiate each term separately
- The derivative of cotangent x is negative cosecant squared x
- The derivative of tangent x is secant squared x
- Multiply by the constants and combine

Answer: $y' = -\csc^2 x + 7 \sec^2 x$

7. Find the derivative using the quotient rule.

$$y = \frac{\cos x}{x}$$

- Identify the numerator as cosine x and the denominator as x
- Apply the quotient rule formula
- The derivative of cosine x is negative sine x and the derivative of x is one
- Simplify the numerator by combining like terms

Answer: $y' = \frac{-x \sin x - \cos x}{x^2}$

8. Find the derivative using the product rule.

$$y = \sin x \cos x$$

- Identify the two factors as sine x and cosine x
- Apply the product rule formula
- The derivative of sine x is cosine x and the derivative of cosine x is negative sine x
- Combine the terms to form the difference of squares

Answer: $y' = \cos^2 x - \sin^2 x$

9. Find the derivative of the function.

$$y = 4 \csc x - 2 \cot x$$

- Differentiate each term using the constant multiple rule
- The derivative of cosecant x is negative cosecant x times cotangent x
- The derivative of cotangent x is negative cosecant squared x
- Apply the negative sign from the second term and combine

Answer: $y' = -4 \csc x \cot x + 2 \csc^2 x$



10. Find the derivative using the product rule.

$$y = e^x \sin x$$

→ Identify the two factors as e to the x and sine x

→ Apply the product rule formula

→ The derivative of e to the x is e to the x and the derivative of sine x is cosine x

→ Factor out the greatest common factor e to the x

Answer: $y' = e^x(\sin x + \cos x)$

